TOPIC 7

Add Multi-Digit Whole Numbers

OVERVIEW

Topic 7 focuses on adding multi-digit whole numbers using mental math, estimation, and a standard algorithm with procedural fluency.

FIND SUMS

Use Mental Math to Add Your child will use the break-apart and make ten strategies to find sums using mental math.

Here is how your child might find $325 + 114$ using the make ten strategy.

Break apart 114.
$114 = 5 + 100 + 9$

Add 5 to 325 to make a ten.
$325 + 5 = 330$

Then add 100.
$330 + 100 = 430$

Finally, add 9.
$430 + 9 = 439$

ESTIMATE SUMS

Use Rounding and Compatible Numbers to Add Your child will learn two ways to estimate sums: rounding and compatible numbers. Compatible numbers are numbers that are close to the original numbers, but are easy to add mentally. Regardless of the estimation method, your child should begin to understand that the closer the replaced numbers are to the original numbers, the closer the estimate will be to the actual sum or difference.

Here are two ways your child might solve the problem "Do the two pandas together weigh more than 500 pounds?"



One Way
Round to the nearest hundred and add.

$$\begin{array}{r} 255 \longrightarrow 300 \\ +\ 329 \longrightarrow 300 \\ \hline 600 \end{array}$$

$255 + 329$ is about 600.
$600 > 500$

So, the pandas together weigh more than 500 pounds.

Another Way

Use compatible numbers to add.

$$\begin{array}{r} 255 \longrightarrow 250 \\ +\ 329 \longrightarrow 325 \\ \hline 575 \end{array}$$

$255 + 329$ is about 575.

$575 > 500$

So, the pandas together weigh more than 500 pounds.

STANDARD ADDITION ALGORITHMS

Record Partial Sums to Add Your child will use place value to break apart numbers and record partial sums to add. There are two ways to use partial sums — starting from the left or starting with the ones.

Here are two ways to find $243 + 179$.

Add each place value. Start with hundreds.

$$\begin{array}{rl} 243 & \\ +\ 179 & \\ \hline 300 & \text{2 hundreds + 1 hundred} \\ 110 & \text{4 tens + 7 tens} \\ +\ \ 12 & \text{3 ones + 9 ones} \\ \hline 422 & \end{array}$$

300, 110, and 12 are partial sums.

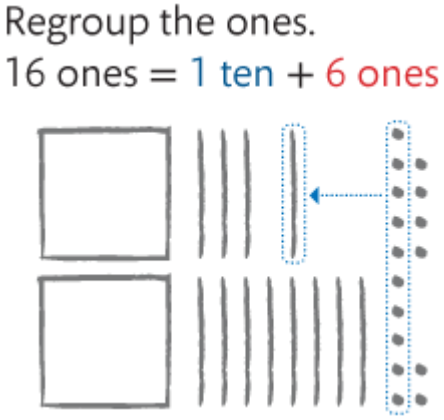
Add each place value. Start with ones.

$$\begin{array}{r} 243 \\ +\ 179 \\ \hline 12 \\ 110 \\ +\ 300 \\ \hline 422 \end{array}$$

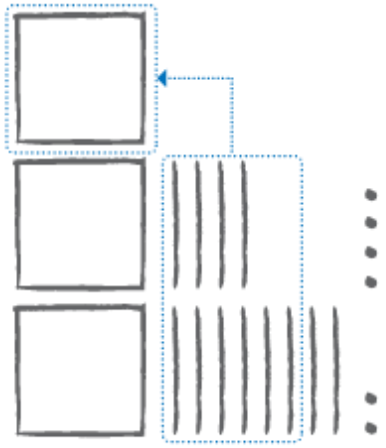
12, 110, and 300 are partial sums.

Regroup Partial Sums Your child will learn to regroup when using partial sums. This will enable your child to learn and use a standard algorithm with understanding.

Here is how your child might find $139 + 187$. Notice how the regrouping is recorded for each step.



Regroup the tens.
12 tens = 1 hundred + 2 tens



Hundreds	Tens	Ones
1	3	9
+ 1	8	7
2	11	~~16~~
2	~~12~~	6
3	2	6

Use a Traditional Algorithm Your child will also add numbers using a traditional algorithm, which records regrouping above the addends.

$$\begin{array}{r} {\scriptstyle 1\,1} \\ 3,785 \\ +\ 2,125 \\ \hline 5,910 \end{array}$$

SOLVE REAL-WORLD PROBLEMS

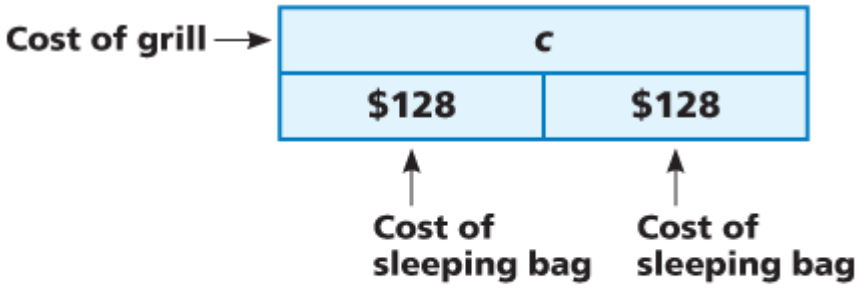
Solve Multi-Digit Comparison Problems Your child will apply the addition methods practiced in this topic, as well as other operations used in earlier topics, to solve real-world comparison problems. For example, "How can Jin compare the cost of the grill to the cost of the sleeping bag?"

DATA

Item	Price
Grill	$256
Backpack	$89
Lantern	$78
Sleeping Bag	$128

One Way

Use addition to compare.

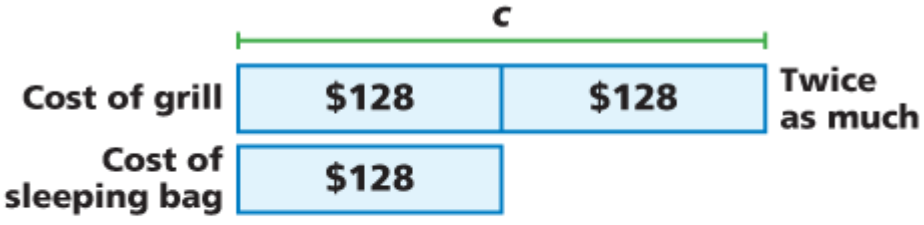


$$c = \$128 + \$128$$
$$c = \$256$$

The cost of the grill is $\$128$ more than the cost of the sleeping bag.

Another Way

Use multiplication to compare.



$$c = 2 \times \$128$$
$$c = \$256$$

The cost of the grill is twice as much as the cost of the sleeping bag.

CONNECT THE MATH

You can connect the math in this topic to everyday experiences. Be aware of situations in which you would add two numbers (up to 4 digits), and enlist your child's help. You might have your child find the sum of two bills, two distances you will drive, or two weights. Help your child determine which method to use — mental math, estimation, or an algorithm. Encourage your child to use a method that can be used reliably and accurately.

TOPIC 7 LESSONS

Lesson 7-1	Mental Math: Addition
Lesson 7-2	Estimate Sums
Lesson 7-3	Use Partial Sums to Add
Lesson 7-4	More Partial Sums: Regrouping
Lesson 7-5	Add Whole Numbers
Lesson 7-6	Add Greater Numbers
Lesson 7-7	**PROCEDURAL FLUENCY PRACTICE** Use a Standard Algorithm to Add
Lesson 7-8	Solve More Real-World Comparison Problems
Lesson 7-9	**MATH THINKING AND REASONING** Complete Tasks with Mathematical Fluency

LESSON 7-1

Mental Math: Addition

MA.3.NSO.2.1, MA.3.AR.1.2, MTR.2.1, MTR.4.1, MTR.5.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Find each sum.

$624 + 171 = \underline{795}$

$171 = 100 + 70 + \underline{1}$

$624 + 100 = \underline{724}$

$724 + 70 = \underline{794}$

$794 + 1 = \underline{795}$

Break 171 into parts: $100 + 70 + 1$. Then add each part, one at a time: $624 + 100 = 724$, then $724 + 70 = 794$, then $794 + 1 = 795$.

HOME ACTIVITY

Number Cube Addition

MATERIALS

Number Cube, Paper, Pencil

ACTIVITY

Have your child make a 3-digit number by rolling a number cube 3 times. Record the number on your paper. Now you roll the number cube and ask your child to add your number to theirs. Here is an example: Your child rolls 3, 3, 6, you roll 5, and your child adds $336 + 5 = 341$. You roll the number cube again to make a 2-digit number. If you rolled a 5 the first time and a 4 this time, the 2-digit number will be 54. Ask your child to add $336 + 54 = 390$. You roll the number cube a third time to make a 3-digit number. If you roll a 2, ask your child to add $336 + 542 = 878$. Encourage your child to describe the method they use to find this sum. For this example, your child might use place value, first adding hundreds, then tens, then ones: $336 + 500 = 836$, $836 + 40 = 876$, and $876 + 2 = 878$. Repeat with new rolls of the number cube.

LESSON 7-2

Estimate Sums

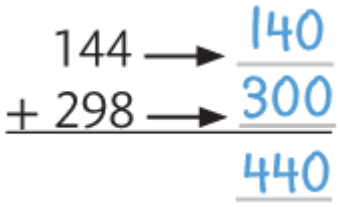
MA.3.NSO.2.1, MA.3.NSO.1.4, MA.3.AR.1.2, MTR.1.1, MTR.4.1, MTR.7.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Round to the nearest ten to estimate.



Round 144 to 140, and round 298 to 300. Add the rounded numbers: $140 + 300 = 440$.

HOME ACTIVITY

What Do You Think?

MATERIALS

Paper, Pencil

ACTIVITY

Write a 3-digit addition problem on your paper, such as $388 + 413$. Say to your child, "I think the sum is 701. What do you think?" The correct sum is 801. Do not tell your child that you stated an incorrect sum. Ask your child to estimate the sum and say if your answer is reasonable. For example, your child might round 388 to 400, round 413 to 400, and say that the sum is about 800. Because the rounded numbers were quite close to the actual numbers, the sum should be close to the estimate. Your sum of 701 is not reasonable, compared to the estimate of 800. Have your child add the numbers to get the correct sum of 801. Repeat several times with different 3-digit addition problems.

LESSON 7-3

Use Partial Sums to Add

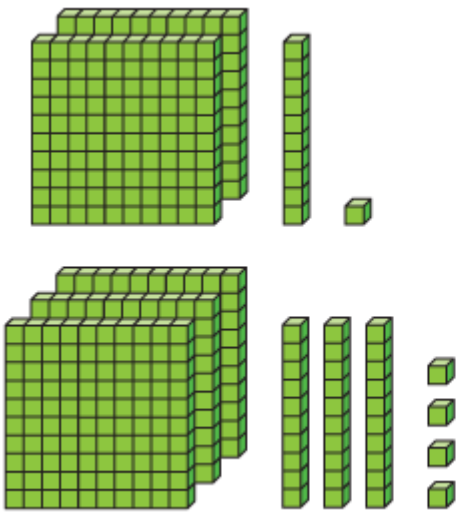
MA.3.NSO.2.1, MA.3.AR.1.2, MTR.2.1, MTR.4.1, MTR.5.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Estimate each sum. Use place-value blocks and partial sums to add.



$$\begin{array}{r} 211 \\ +\ 334 \\ \hline 500 \\ 40 \\ +\ \ 5 \\ \hline 545 \end{array}$$

Estimate: 200 + 300 = 500

To estimate, round each number to the nearest 100 then add. Round 211 to 200, round 334 to 300, and add $200 + 300 = 500$.

To add, use partial sums.
Add hundreds: $200 + 300 = 500$.

Add tens: $10 + 30 = 40$.
Add ones: $1 + 4 = 5$.
Add the partial sums: $500 + 40 + 5 = 545$.

HOME ACTIVITY

Shopping Trip

MATERIALS

Sale Flyer, Paper, Pencil

ACTIVITY

Go on a pretend shopping trip with your child. Find two items in a store sale flyer with prices that can be rounded to a 3-digit number. Assist your child with the rounding. For example, a dishwasher is priced at $\$485.95$ and rounds to $\$486$. A lawn mower is priced at $\$388.88$ and rounds to $\$389$. Ask your child to find the total cost to purchase both the dishwasher and the lawn mower. Have your child describe their method for finding the sum $486 + 389$. They may choose to use partial sums: $400 + 300 = 700$, $80 + 80 = 160$, $6 + 9 = 15$, then the sum is $700 + 160 + 15 = 875$. The two items would cost about $\$875$. Choose other items in the flyer to use to help your child practice adding 3-digit numbers.

LESSON 7-4

More Partial Sums: Regrouping

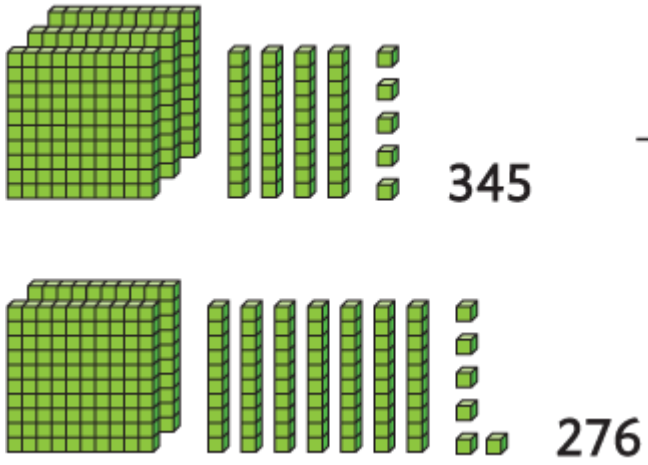
MA.3.NSO.2.1, MA.3.AR.1.2, MTR.2.1, MTR.3.1, MTR.6.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Use the place-value blocks to help add.



$$\begin{array}{r} 345 \\ +\ 276 \\ \hline 621 \end{array}$$

To add, use partial sums and regroup.
Add ones:
5 ones plus 6 ones equal 11 ones. Regroup as 1 ten and 1 one.

Add tens:
1 ten plus 4 tens plus 7 tens equal 12 tens. Regroup as 1 hundred and 2 tens.

Add hundreds:
1 hundred plus 3 hundreds plus 2 hundreds equal 6 hundreds.

Add the partial sums:
6 hundreds plus 2 tens plus 1 one equal 621.

HOME ACTIVITY

1 Number, 3 Ways

MATERIALS

Number Cube, Paper, Pencil

ACTIVITY

Have your child roll a number cube 3 times to make a 3-digit number. Write it on your paper. For example, your child rolls 5, 6, 3, and makes the number 563. Ask your child to write their number in expanded form: $500 + 60 + 3$. Now ask them to regroup some of the tens to write the expanded form a different way, such as: $500 + 50 + 13$. Next ask them to regroup some of the hundreds to write the expanded form a third way, such as: $400 + 160 + 3$. Look at the 3 ways that the number 563 is written: $500 + 60 + 3$, $500 + 50 + 13$, and $400 + 160 + 3$. Have your child add each expression to verify that it equals 563. Repeat several times with other 3-digit numbers.

LESSON 7-5

Add Whole Numbers

MA.3.NSO.2.1, MA.3.AR.1.2, MTR.2.1, MTR.4.1, MTR.5.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Find the sum.

$$\begin{array}{r} 456 \\ +\,134 \\ \hline 590 \end{array}$$

To add, use regrouping.
Add ones:
6 ones plus 4 ones equal 10 ones. Regroup as 1 ten and 0 ones.

Add tens:
1 ten plus 5 tens plus 3 tens equal 9 tens.

Add hundreds:
4 hundreds plus 1 hundred equal 5 hundreds.

HOME ACTIVITY

Pancake Party

MATERIALS

Glass, Scissors, Spatula, Baking Sheet, Paper, Pencil

ACTIVITY

Trace 10 circles with a drinking glass and cut them out to make 10 “pancakes”. Write one 3-digit number on each pancake. Mix up the pancakes and place them face down on a baking sheet. Tell your child that they can choose two pancakes to flip over with the spatula. Then they can use any method they would like to find the sum of the two numbers. Check the answer, and once it is correct your child takes the pancakes off the baking sheet to “eat” them! Have your child continue to flip pancakes and find sums until all the pancakes are gone. Encourage your child to show you different methods for finding the sums, such as place value, partial sums, and regrouping.

LESSON 7-6

Add Greater Numbers

MA.3.NSO.2.1, MA.3.AR.1.2, MTR.1.1, MTR.4.1, MTR.5.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Choose a method to find the sum.

$$\begin{array}{r} 3{,}945 \\ +\ 3{,}343 \\ \hline 7{,}288 \end{array}$$

To add, use regrouping.
Add ones:
5 ones plus 3 ones equal 8 ones.

Add tens:
4 tens plus 4 tens equal 8 tens.

Add hundreds:
9 hundreds plus 3 hundreds equal 12 hundreds. Regroup as 1 thousand and 2 hundreds.

Add thousands:

1 thousand plus 3 thousands plus 3 thousands equal 7 thousands.

HOME ACTIVITY

Numbers in the News

MATERIALS

Newspaper, Paper, Pencil

ACTIVITY

Help your child scan articles in a newspaper or an online news source and find two 4-digit numbers. Ask your child to estimate the sum of these numbers and then use any method to add them. For example: "Summer rentals at the beach have increased 6,430 since last summer." and "A new restaurant is opening at 1124 Broad Boulevard." Have your child describe their method for estimating, such as: 6,430 is about 6,400, and 1,124 is about 1,100, so $6{,}430 + 1{,}124$ is about $6{,}400 + 1{,}100$ or 7,500. Your child might choose to use place value to add the numbers, like this:
$6{,}430 + 1{,}124 = 6{,}000 + 1{,}000 + 400 + 100 + 30 + 20 + 4$ which equals
$7{,}000 + 500 + 50 + 4 = 7{,}554$. Ask your child if this answer is reasonable based on the estimate. Repeat with other 4-digit numbers you find in the news.

LESSON 7-7

Procedural Fluency Practice: Use a Standard Algorithm to Add

MA.3.NSO.2.1, MA.3.AR.1.2, MTR.1.1, MTR.3.1, MTR.6.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Choose any method to find the sum. Then use estimation to see if your answer is reasonable.

$$\begin{array}{r} 518 \\ +\ 339 \\ \hline 857 \end{array}$$

To add, use regrouping.
Add ones:
8 ones plus 9 ones equal 17 ones. Regroup as 1 ten and 7 ones.

Add tens:
1 ten plus 1 ten plus 3 tens equal 5 tens.

Add hundreds:
5 hundreds plus 3 hundreds equal 8 hundreds.

To estimate, round each addend to the nearest 10 and add the rounded numbers. 518 rounds to 520, 339 rounds to 340, and $520 + 340 = 860$.

The answer of 857 is reasonable because it is close to the estimate of 860.

HOME ACTIVITY

Roll 2, Roll 3, Roll 4

MATERIALS

Number Cube, Paper, Pencil

ACTIVITY

Help your child practice using a standard algorithm to add. Ask your child to roll a number cube 2 times to make a 2-digit number, then roll it 2 more times to make another 2-digit number. Use the numbers to write down an addition problem, such as: $36 + 45$. Have your child add the two numbers using a standard algorithm. Have them check their work by using place value or partial sums. Now have your child roll the number cube again to add a place to their first number. For example, a roll of 4 will change 36 to 364. Roll again to add a place to the second number. For example, a roll of 2 will change 45 to 452. Ask your child to find the sum of these 3-digit numbers: $364 + 452$. Repeat one more time by rolling the number cube to add a place to each number to make two 4-digit numbers to add.

LESSON 7-8

Solve More Real-World Comparison Problems

MA.3.AR.1.2, MA.3.AR.2.3, MTR.3.1, MTR.6.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Solve the problem.

Lupita cleans 7 times as many paint brushes as Mike. Mike cleans 8 paint brushes. How many paint brushes does Lupita clean?

56 paint brushes

Lupita cleans more brushes than Mike does, 7 times as many. Multiply 7 by the number of brushes that Mike cleans. Mike cleans 8 brushes: $7 \times 8 = 56$.

HOME ACTIVITY

Bank Account

MATERIALS

Paper, Pencil

ACTIVITY

Use a piece of paper to create a pretend bank account statement. Write, "Balance: $ xxx" at the top. On the next line write, "Withdrawal: $\$265$". Below that write, "Balance: $\$124$". Tell your child that the balance is the total amount of money in the bank account. A withdrawal is an amount of money you take out of the account. A deposit is an amount of money you add to the account. Ask your child how they would find how much money was in this bank account before you made the withdrawal. They will add $265 + 124$, then estimate to check the answer, and replace "xxx" at the top with 389. Continue this activity by writing "Deposit: $\$319$" on the next line and "Balance: $xxx" on the following line. Ask your child to find the account balance after the deposit of $\$319$ was made.